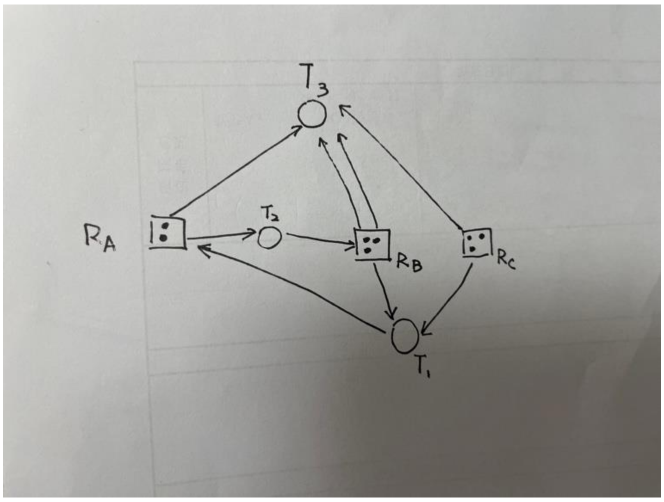
Exercise 3 Solutions

1. Resource allocation graph is specified below:



Here is a cycle in this resource allocation graph, but the system is in a safe condition. The process 3 can be satisfied without waiting for other resources. After process 3 releases its resources, the process 1 and process 2 can be satisfied.

2. (a) A-9; B-13; C-10; D-11

(b) Need[i, j]=Max[i,j]-Allocation[i,j] so content of Need matrix is ABCD

P0 2 0 1 1
P1 0 6 5 0
P2 1 1 0 2
P3 1 0 2 0
P4 1 4 4 4

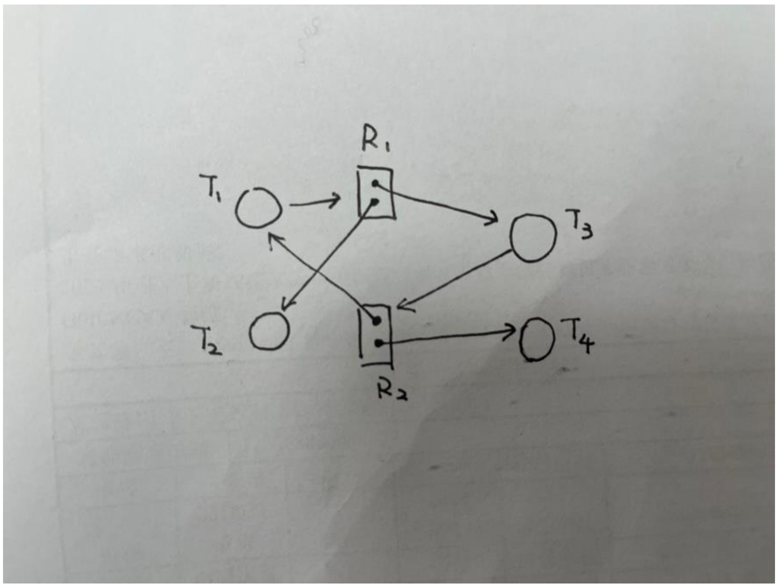
(c) The system is in a safe state as the processes can be finished in the sequence P0, P2, P3, P4 and P1. (P0, P2, P3, P1 and P4)

(d) If a request from process P4 arrives for additional resources of (1,2,0,0,), and if this request is granted, the new system state would be tabulated as follows.

	Max				Allocation				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	2	0	1	1				
P1	1	7	5	0	1	1	0	0	0	6	5	0				
P2	2	3	5	6	1	2	5	4	1	1	0	2				
P3	1	6	5	3	0	6	3	3	1	0	2	0				
P4	1	6	5	6	1	4	1	2	0	2	4	4				
													2	0	1	1

After P0 completes, we have (6,0,1,2). There is no one can be finished. So the new system is not at a safe sequence.

3. a. Draw the resource allocation graph.



b. T1 R1 T3 R2 T1

c. No. There is a cycle, but no deadlock. T2 and T4 have all resources for completing. T2 T4, T1, T3

4. The need matrix is as follows:

	R1	R2	R3	R4	R5
A	0	1	0	0	2
B	0	2	1	0	0
C	1	0	3	0	0
D	0	0	1	1	1

Suppose that we are in a safe state. Process D must run first, because we have no other choice. To make process D run, the number X of R4 should be no less than 1. Since process A, B, and C do not need any more instance of resource R4, the constraint of X is X>=1.

So if and only if X>=1, the state is safe. Then the smallest value of X is 1.